

Tugas Week 13
Jaringan Komputer



Nama Dan Nim:
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Disini kita akan protokol Ethernet dan protokol ARP. RFC 826 (<ftp://ftp.rfc-editor.org/in-notes/std/std37.txt>) berisi detail mengenai dari protokol ARP, yang digunakan oleh perangkat IP untuk menentukan alamat IP dari antarmuka jarak jauh yang alamat Ethernetnya diketahui.

Menangkap dan menganalisis frame Ethernet

1. <http://gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html>

THE BILL OF RIGHTS <i>Amendments 1-10 of the Constitution</i> The Conventions of a number of the States having, at the time of adopting the Constitution, expressed a desire, in order to prevent misconstruction or abuse of its powers, that further declaratory and restrictive clauses should be added, and as extending the ground of public confidence in the Government will best insure the beneficent ends of its institution; Resolved, by the Senate and House of Representatives of the United States of America, in Congress assembled, two-thirds of both Houses concurring, that the following articles be proposed to the Legislatures of the several States, as amendments to the Constitution of the United States; all or any of which articles, when ratified by three-fourths of the said Legislatures, to be valid to all intents and purposes as part of the said Constitution, namely: Amendment I Congress shall make no law respecting an establishment of religion, or prohibiting the free exercise thereof; or abridging the freedom of speech, or of the press; or the right of the people peaceably to assemble, and to petition the government for a redress of grievances. Amendment II A well regulated militia, being necessary to the security of a free state, the right of the people to keep and bear arms, shall not be infringed. Amendment III No soldier shall, in time of peace be quartered in any house, without the consent of the owner, nor in time of war, but in a manner to be prescribed by law. Amendment IV The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated, and no warrants shall issue, but upon probable cause, supported by oath or affirmation, and particularly describing the place to be searched, and the persons or things to be seized. Amendment V No person shall be held to answer for a capital, or otherwise infamous crime, unless on a presentment or indictment of a grand jury, except in cases arising in the land or naval forces, or in the militia, when in actual service in time of war or public danger; nor shall any person be subject for the same offense to be twice put in jeopardy of life or limb; nor shall be compelled in any criminal case to be a witness against himself, nor be deprived of life, liberty, or property, without due process of law; nor shall private property be taken for public use, without just compensation. Amendment VI In all criminal prosecutions, the accused shall enjoy the right to a speedy and public trial, by an impartial jury of the state and district wherein the crime shall have been committed, which district shall have been previously ascertained by law, and to be informed of the nature and cause of the accusation; to be confronted with the witnesses against him; to have compulsory process for obtaining witnesses in his favor, and to have the assistance of counsel for his defense.	
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2. HTTP after open gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html

Wireshark packet capture showing HTTP traffic. The packet list shows a GET request for /gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html. The packet details pane shows the request structure, including the Host, User-Agent, and Accept headers. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
104	3.601638900	192.168.0.102	192.168.0.1	HTTP	250	GET /gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html HTTP/1.1
109	3.626818800	192.168.0.1	192.168.0.102	HTTP	513	HTTP/1.1 200 OK
561	12.1355330	192.168.0.102	128.119.245.12	HTTP	550	GET /gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html HTTP/1.1
584	12.4275220	128.119.245.12	192.168.0.102	HTTP	562	HTTP/1.1 200 OK (text/html)
588	12.4891044	192.168.0.102	128.119.245.12	HTTP	496	GET /favicon.ico HTTP/1.1
596	13.5706501	128.119.245.12	192.168.0.102	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)

Frame 561: Packet, 550 bytes on wire (4400 bits), 550 bytes captured (4400 bits) on interface 0

Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink_5b:1b:b4 (bc:07:1d:5b:1b:b4)

Internet Protocol Version 4, Src: 192.168.0.102, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 63013, Dst Port: 80, Seq: 1, Ack: 1, Len: 496

Hypertext Transfer Protocol

GET /gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html HTTP/1.1

Host: gaia.cs.umass.edu

Connection: keep-alive

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/109.0.0.0 Safari/537.36

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/svg+xml,*/*;q=0.8

Accept-Encoding: gzip, deflate

Accept-Language: en-US,en;q=0.9,en-GB;q=0.8,en;q=0.7

[Response in frame 584]

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html]

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Host: gaia.cs.umass.edu

Connection: keep-alive

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/109.0.0.0 Safari/537.36

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/svg+xml,*/*;q=0.8

Accept-Encoding: gzip, deflate

Accept-Language: en-US,en;q=0.9,en-GB;q=0.8,en;q=0.7

[Response in frame 584]

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html]

Address Resolution Protocol

Caching ARP

1. CMD arp -a

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz>arp -a

Interface: 192.168.0.102 --- 0xb
Internet Address      Physical Address      Type
192.168.0.1           bc-07-1d-5b-1b-b4     dynamic
192.168.0.255         ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\fairuuz>
```

2. paket ARP

The image shows a Wireshark packet capture of an ARP request and response. The packet list on the left shows a packet at time 33.113525907 seconds, source TPLink_5b:1b:b4, destination LiteonTechno_af:cf:7d, protocol ARP, length 42 bytes. The packet details pane shows the Ethernet II header with source TPLink_5b:1b:b4 and destination LiteonTechno_af:cf:7d, and the ARP request details. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
33	11.3525907...	TPLink_5b:1b:b4	LiteonTechno_af:cf:7d	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
34	11.3526305...	LiteonTechno_af:cf:7d	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
212	30.6122030...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
219	32.5578756...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
250	43.4185193...	TPLink_5b:1b:b4	LiteonTechno_af:cf:7d	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
251	43.4185404...	LiteonTechno_af:cf:7d	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
332	52.4241806...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
333	53.4487294...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
337	54.4734655...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
500	94.7230723...	TPLink_5b:1b:b4	LiteonTechno_af:cf:7d	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
501	94.7231154...	LiteonTechno_af:cf:7d	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
717	124.384280...	TPLink_5b:1b:b4	LiteonTechno_af:cf:7d	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
718	124.384306...	LiteonTechno_af:cf:7d	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
845	164.257085...	TPLink_5b:1b:b4	LiteonTechno_af:cf:7d	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
846	164.257081...	LiteonTechno_af:cf:7d	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d
902	177.808469...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.104? Tell 192.168.0.1
936	195.073700...	TPLink_5b:1b:b4	Broadcast	ARP	42	Who has 192.168.0.102? Tell 192.168.0.1
937	195.073718...	LiteonTechno_af:cf:7d	TPLink_5b:1b:b4	ARP	42	192.168.0.102 is at 24:b2:b9:af:cf:7d

Frame 33: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF{...} (0.0.0.0)

Ethernet II, Src: TPLink_5b:1b:b4 (bc:07:1d:5b:1b:b4), Dst: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d)

Address Resolution Protocol (request)

24 b2 b9 af cf 7d bc 07 1d 5b 1b b4 08 06 00 01 00 00 06 04 00 01 bc 07 1d 5b 1b b4 c0 a8 00 01 00 00 00 00 00 00 c0 a8 00 66